

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks. We propose the Transformer, a model architecture eschewing recurrence and entirely relying on an attention mechanism to draw global dependencies between input and output.

An attention function can be described as mapping a query and a set of key-value pairs to an output. The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key.

Scaled Dot-Product Attention: We compute the attention function on a set of queries simultaneously, packed together into a matrix Q . The keys and values are also packed into matrices K and V . We compute the matrix of outputs as $\text{softmax}(QK^T / \sqrt{d_k})V$.

Multi-Head Attention: Instead of performing a single attention function with d_{model} -dimensional keys, values and queries, we found it beneficial to linearly project the queries, keys and values h times with different, learned linear projections.

Positional Encoding: Since our model contains no recurrence and no convolution, in order for the model to make use of the order of the sequence, we must inject some information about the relative or absolute position of the tokens.

On the WMT 2014 English-to-German translation task, the big transformer model achieves a state-of-the-art BLEU score of 28.4, outperforming the best existing models including ensembles.